

Mayurdeep Pathak

M.Tech (Research), Indian Institute of Science
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EDUCATION

- **M.Tech (Research) - Indian Institute of Science** Aug. 2023 – Sep. 2026 (Expected)
Dept. of Electrical Communication Engineering
– Advisor: Prof. Rajiv Soundararajan, Visual Information Processing Lab Bangalore, India
- **B.E. - Jorhat Engineering College** Aug 2016 – July 2020
Dept. of Computer Science and Engineering Jorhat, India

RESEARCH INTEREST

3D Vision, Dynamic Scene Reconstruction, Novel View Synthesis, Image and Video Processing

PUBLICATIONS

- **Sustaining Reconstruction Quality in Long-Duration Dynamic 3D Gaussian Splatting**
Mayurdeep Pathak, Vikram Sandu, Rajiv Soundararajan Project page
- **Temporally Aware Densification for Dynamic 3D Gaussian Splatting** ECCV 2026
Vikram Sandu, Mayurdeep Pathak, Rajiv Soundararajan Project page · arXiv
- **GlobeMetrics: A Healthcare Framework for Video Based Saccade Characterization** IEEE CAI 2024
Debanga Raj Neog, Pharvesh Salman Choudhary, Mayurdeep Pathak IEEE Xplore

RESEARCH EXPERIENCE

- **Visual Information Processing Lab, IISc** Aug 2023 - Present
M.Tech (Research) Student
I work on **dynamic 3D Gaussian Splatting** for novel view synthesis of multi-view video. My focus is improving reconstruction quality in novel views, in particular in the moving regions where current methods blur. The two works below first improve quality on short sequences, then extend those gains to long-duration video.
 - **Temporally Aware Densification for Dynamic 3DGS** (co-author, ECCV 2026). Given multiple video cameras recording a moving scene, existing Dynamic 3DGS methods reconstruct moving objects blurrily. We identified the flaw. These methods carry over a rule for adding model capacity that was designed for static scenes, so it adds capacity everywhere except where the motion actually is. We fixed this by making that process aware of motion.
 - **Sustaining Reconstruction Quality in Long-Duration Dynamic 3DGS** (first author). The above works well on short clips of a few seconds, but real multi-view video runs for minutes, and quality collapses in the moving regions well before then. We identify the root cause and solve it, giving significant gains across several datasets.

WORK EXPERIENCE

- **Junior Research Fellow**, Mehta Family School of Data Science and AI, IIT Guwahati Mar 2023 – Jul 2023
- **Data Engineer**, Itonim Analytics May 2021 – Mar 2022
- **Software Engineer Consultant**, Five Vectors Inc. Sep 2020 – Feb 2021

POSITIONS OF RESPONSIBILITY

- **Secretary**, IISc ACM Student Chapter, Indian Institute of Science 2025 - 2026

TEST SCORES

GATE 2023 - CS-IT - All India Rank 662